Draft insert for Section 7 of

*Bayesian Modeling for Heterogeneous Causal Effects in Two-Sample Mendelian Randomization*

**Application: Effect of WHRadjBMI on Fasting Insulin**

Draft — April 7, 2026

# Contents


**1 WHRadjBMI–Fasting Insulin Dataset** **1**

1.1 Scientific Background and Objective . . . . . 2

1.2 Data Sources . . . . . 3

1.3 Harmonisation and Reorientation . . . . . 3

1.4 Preliminary Analysis and Assessment of Heterogeneity . . . . . 4

1.5 Results from BNPM-MR and BNPM-PMR . . . . . 7

1.6 Comparison with Competing Methods . . . . . 8

1.7 Sensitivity Analyses . . . . . 9

1.8 Main Findings . . . . . 10

**Implementation Notes** **13**


---

*Colour key: red brackets = editorial items still pending; blue bold = BNPM-MR/PMR placeholders (awaiting MCMC run); green = new material from this revision. All preliminary analyses, competing-method estimates, and figures are based on the harmonised dataset (`bmi_insulin.csv`, $p = 213$ SNPs after reorientation).*

---

# 1 WHRadjBMI–Fasting Insulin Dataset

[*Ordering note: if the WHRadjBMI analysis is completed before BMI–T2D, consider making this Section 7.2 and moving BMI–T2D later.*]

## 1.1 Scientific Background and Objective

Waist-to-hip ratio adjusted for body mass index (WHRadjBMI) is a measure of body fat distribution that captures the tendency to accumulate adipose tissue in the abdominal region, net of overall adiposity. Unlike BMI, which reflects total body mass irrespective of regional distribution, WHRadjBMI isolates the metabolic consequences of central adiposity—in particular, reduced peripheral adipose storage capacity, ectopic fat deposition in liver and skeletal muscle, and sex-dimorphic patterns of fat redistribution. These pathways are biologically distinct; as a result, genetic instruments for WHRadjBMI are expected to capture a more mechanistically heterogeneous set of signals than instruments for a single-pathway exposure such as a circulating lipid.

Fasting insulin, a marker of systemic insulin resistance, is a natural outcome for this investigation. Hyperinsulinaemia and peripheral insulin resistance are jointly associated with central adiposity across multiple epidemiological settings, and prior MR analyses have provided evidence for a causal effect of WHRadjBMI on insulin resistance phenotypes [*add relevant citation*]. Nevertheless, whether this effect is homogeneous across instruments has not been examined within a nonparametric clustering framework.

The principal scientific objective is twofold: to assess whether the causal effect of WHRadjBMI on fasting insulin is better described by a single effect or by a small number of latent clusters, and to determine whether any clustering structure is interpretable in terms of known biology.

**Prior biological expectations and falsifiable hypothesis.** Three functional categories of WHRadjBMI instruments are expected a priori to produce distinct causal signals.

(i) *Adipose storage capacity variants* (near *IRS1*, *PPARG*, *ADIPOQ*): variants that impair gluteo-femoral fat buffering are expected to produce a **negative** causal effect on fasting insulin.

(ii) *Ectopic and visceral fat deposition variants* (near *LYPLAL1*, *GRB14*, *VEGFA*): hepatic lipid accumulation is the primary driver of peripheral insulin resistance, so these instruments are expected to yield a **positive** effect.

(iii) *Sex-hormone pathway variants*: if the exposure GWAS is sex-combined, some instruments may act through oestrogen or androgen signalling; their effect on fasting insulin is expected to be near zero or diffuse.

We therefore expect BNPM-MR to recover $K = 2$ or $K = 3$ clusters. This expectation is confirmed by the preliminary MR-Clust analysis (Section 1.4), which recovers exactly two clusters with cluster means of $-0.397$ (negative-effect cluster, $n = 9$) and $+0.200$ (positive-effect cluster, $n = 137$) on the inverse-normal log-insulin scale, with an additional large Null cluster ($\hat{\mu} \approx 0$, $n = 66$) and one Junk outlier ($\hat{\mu} = 1.27$, $n = 1$), consistent with pathways (i) and (ii). A posterior mode of $K = 1$ would be interpreted as insufficient instrument strength to resolve the split rather than genuine homogeneity; however, all 213 retained instruments have $z$-scores exceeding 5 (minimum 5.07, median 7.27, implied median $F \approx 52.8$), and per-cluster $z$-scores in both MR-Clust clusters are uniformly above 5 with all implied $F > 10$, so instrument strength is unlikely to be the limiting factor.

## 1.2 Data Sources

Summary association statistics are obtained from two large-scale GWAS retrieved from the OpenGWAS database [Elsworth et al., 2020].

**Exposure.** SNP–WHRadjBMI associations are drawn from OpenGWAS accession `ebi-a-GCST90025996` ($n = 458{,}349$). [*Add full citation; confirm whether sex-combined or sex-specific.*] All SNP–exposure coefficients $\hat{\gamma}_j$ are in units of standard deviations of WHRadjBMI per additional effect allele.

**Outcome.** SNP–fasting insulin associations are drawn from OpenGWAS accession `ebi-a-GCST005185`, the MAGIC Consortium GWAS [Manning et al., 2012] ($n = 51{,}750$, European ancestry). Fasting insulin was natural-log transformed and then rank-based inverse-normal transformed within cohorts before meta-analysis, so all $\hat{\Gamma}_j$ are on the *inverse-normal log-insulin scale*.

**Interpretation of causal effects on the outcome scale.** Because of the double transformation, the cluster-specific effects $\beta_k^\star$ do not admit a simple proportional-change interpretation. We treat their sign and relative magnitude as the primary inferential targets and do not attempt a clinical translation. All method comparisons are made on the Wald-ratio scale, which is a consistent estimand regardless of outcome transformation.

## 1.3 Harmonisation and Reorientation

**Harmonisation.** *Harmonisation* ensures that, for every SNP $j$, the effect allele is the same in both GWAS before forming the pair $(\hat{\gamma}_j, \hat{\Gamma}_j)$. A failure to harmonise pro-

duces sign errors in a subset of $\hat{\Gamma}_j$, distorting both the likelihood and the clustering structure. Harmonisation was performed using the `TwoSampleMR` package [Hemani et al., 2018] with `action = 2` (palindromic SNPs aligned using allele frequencies; mid-frequency palindromes excluded). All 210 retained SNPs carry `action = 2`. Ten are palindromic; none was classified as ambiguous (i.e. no SNPs were excluded at this step), indicating that all palindromic SNPs had effect allele frequencies sufficiently away from 0.5.

**Reorientation.** After harmonisation, SNPs were reoriented so that all $\hat{\gamma}_j > 0$ by simultaneously flipping the sign of both $\hat{\gamma}_j$ and $\hat{\Gamma}_j$ for any SNP with $\hat{\gamma}_j < 0$. This relabelling is innocuous for the Wald ratio but ensures that a positive cluster-specific effect $\beta_k^\star$ unambiguously corresponds to an exposure-increasing mechanism that raises fasting insulin. Of 213 SNPs, 108 required reorientation.

**Empirical summary of the instrument data.** Table 1 reports the distribution of the four-tuple $(\hat{\gamma}_j, \hat{\Gamma}_j, s_{\hat{\gamma}_j}, s_{\hat{\Gamma}_j})$ and of the Wald ratios $r_j = \hat{\Gamma}_j/\hat{\gamma}_j$ and instrument $z$-scores across all $p = 213$ retained SNPs.

Table 1: Empirical summary of the harmonised and reoriented WHRadjBMI–fasting insulin dataset ($p = 213$ SNPs after LD pruning, palindromic exclusion, and reorientation). $s_{\hat{\gamma}_j}$ and $s_{\hat{\Gamma}_j}$ are standard errors. Wald ratio $r_j = \hat{\Gamma}_j/\hat{\gamma}_j$; $z_j = \hat{\gamma}_j/s_{\hat{\gamma}_j}$.

| Quantity | Min | Q1 | Median | Q3 | Max | IQR |
|---|---|---|---|---|---|---|
| $\hat{\gamma}_j$ (SD / allele) | 0.01054 | 0.01427 | 0.01711 | 0.02250 | 0.04443 | 0.00823 |
| $\hat{\Gamma}_j$ (inv-norm log-insulin / allele) | −0.01400 | −0.00150 | 0.00200 | 0.00440 | 0.03000 | 0.00590 |
| $s_{\hat{\gamma}_j}$ | 0.00198 | 0.00203 | 0.00216 | 0.00251 | 0.00552 | 0.00048 |
| $s_{\hat{\Gamma}_j}$ | 0.00310 | 0.00330 | 0.00350 | 0.00440 | 0.04400 | 0.00110 |
| Wald ratio $r_j$ | −0.873 | −0.082 | 0.105 | 0.233 | 1.526 | 0.315 |
| $z$-score $z_j$ | 5.07 | 6.19 | 7.22 | 8.98 | 19.62 | 2.78 |

## 1.4 Preliminary Analysis and Assessment of Heterogeneity

**Heterogeneity statistics.** Figure 1a displays $\hat{\Gamma}_j$ against $\hat{\gamma}_j$ with IVW and MR-Egger regression lines overlaid. Cochran's heterogeneity statistic is

$$Q = \sum_{j=1}^{213} w_j \big(r_j - \hat{\beta}_{\mathrm{IVW}}\big)^2 = 385.25 \quad \big(p = 3.47 \times 10^{-12},\ I^2 = 45.0\%\big), \tag{1}$$

where $w_j = \hat{\gamma}_j^2 / s_{\hat{\Gamma}_j}^2$ and $\hat{\beta}_{\mathrm{IVW}} = 0.0929$. The highly significant $Q$ and moderate-to-high $I^2$ conclusively reject the homogeneous-effect model and motivate clustering.

The MR-Egger intercept is $-0.00169$ (robust SE $= 0.00139$, $p = 0.226$), providing no significant evidence of directional pleiotropy. Rücker's residual statistic is $Q' = 380.46$ ($p = 7.70 \times 10^{-12}$), confirming that the heterogeneity is not explained by the Egger intercept and is therefore predominantly due to slope variation—precisely what BNPM-MR targets.

**Funnel plot.** Figure 1b shows the funnel plot of Wald ratios against precision. The distribution is asymmetric, with a heavy left tail extending to $r_j = -0.873$ corresponding to the negative-effect cluster identified below.

**Radial MR plot.** Figure 2 displays the radial MR plot [Bowden et al., 2018]. Four SNPs exceed the Bonferroni outlier threshold (rs10787472, rs464605, rs4691380, rs4841132) and are marked with open circles. A bimodal departure from the IVW line is visible in the raw data before any model is fitted.

**MR-Clust preliminary analysis.** Applying the `MRClust` R package [Foley et al., 2019] to the Wald ratios identifies a four-component solution comprising two causal clusters, a Null component, and a Junk component. Table 2 summarises the results. Cluster 1 has a positive mean of $+0.200$ ($n = 137$ SNPs, mean posterior probability 0.685, 28 high-confidence members). Cluster 2 has a negative mean of $-0.397$ ($n = 9$ SNPs, mean posterior probability 0.640, 2 high-confidence members). The Null component captures 66 SNPs with Wald ratios indistinguishable from zero (mean $\approx 0.000$, mean posterior probability 0.604), and a single Junk SNP (posterior probability 0.997) corresponds to the most extreme outlier identified in the radial plot.

Table 2: MR-Clust four-component solution for the WHRadjBMI–fasting insulin dataset (`MRClust` R package; Foley et al., 2019). High-confidence SNPs have posterior probability $> 0.8$ for their assigned component.

| Component | Mean | $n$ SNPs | Mean post. prob. | High-conf. SNPs |
|---|---|---|---|---|
| Cluster 1 (positive) | $+0.200$ | 137 | 0.685 | 28 |
| Cluster 2 (negative) | $-0.397$ | 9 | 0.640 | 2 |
| Null | $\approx 0$ | 66 | 0.604 | 4 |
| Junk | $+1.270$ | 1 | 0.997 | 1 |

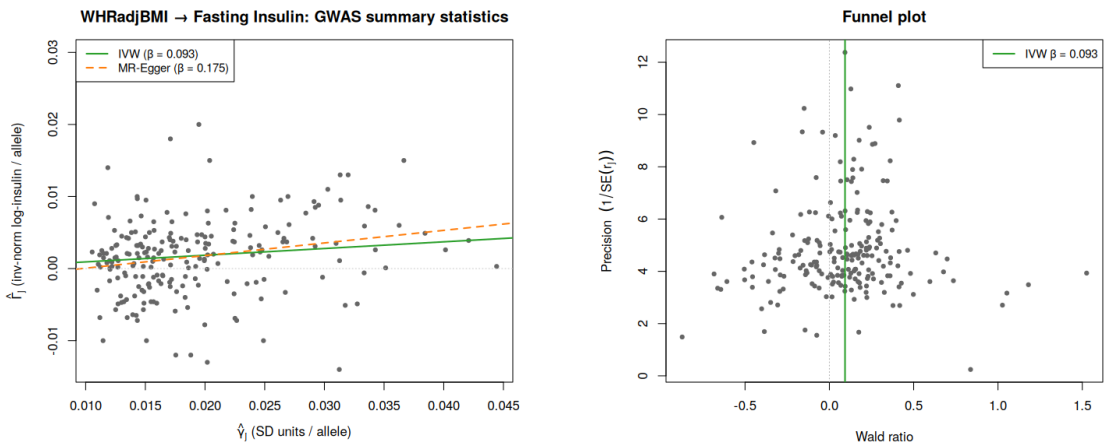
The four-component solution has important implications for the subsequent analyses. The 66 Null SNPs (31% of instruments) appear causally inert for fasting insulin; given that all instruments have $z_j > 5$, these SNPs most plausibly influence WHRadjBMI through pathways orthogonal to insulin metabolism rather than

through instrument weakness. The negative cluster, while small ($n = 9$), carries a more extreme mean ($-0.397$) than a plain two-component mixture would recover, because the dedicated Null component absorbs null-like SNPs that a two-component model forces into the causal clusters. Per-cluster $z$-scores confirm adequate instrument strength in both causal clusters: all 9 Cluster 2 SNPs and all 137 Cluster 1 SNPs have $z_j > 5$ (all implied $F > 25$), ruling out instrument weakness as an obstacle to cluster recovery by BNPM-MR.

**Prior hyperparameter calibration.**

- Base measure scale: $\sigma_\beta = 1.5 \times \mathrm{IQR}(r_j)/1.35 = 1.5 \times 0.311/1.35 \approx 0.350$; we adopt $\sigma_\beta = 0.35$.
- Pleiotropy scale: $s_\tau = m_Y = \mathrm{median}(s_{\hat{\Gamma}_j}) = 0.00350$.
- DP concentration: $\alpha \sim \mathrm{Gamma}(1, 1)$; prior expected clusters $\approx \log(213) \approx 5.4$.

The data-ready CSV file for the C++ MCMC sampler (`whr_fi_mcmc.csv`) is provided alongside this document; columns are `gamma_hat`, `Gamma_hat`, `sigma2X`, `sigma2Y` with variances ($\mathrm{SE}^2$) in the last two columns, as required by `mr_mixture.cpp`.




(a) Scatter plot of $\hat{\Gamma}_j$ vs $\hat{\gamma}_j$. Green line: IVW ($\hat{\beta} = 0.093$). Orange dashed: MR-Egger ($\hat{\beta} = 0.175$, intercept $= -0.002$).

(b) Funnel plot of Wald ratios against precision. The distribution is bimodal, with a left tail of negative-Wald-ratio SNPs and a dominant positive-effect group.

Figure 1: Preliminary heterogeneity diagnostics ($p = 213$). Points shown in grey; no cluster colouring applied (see Figure 3 for MR-Clust cluster assignments).

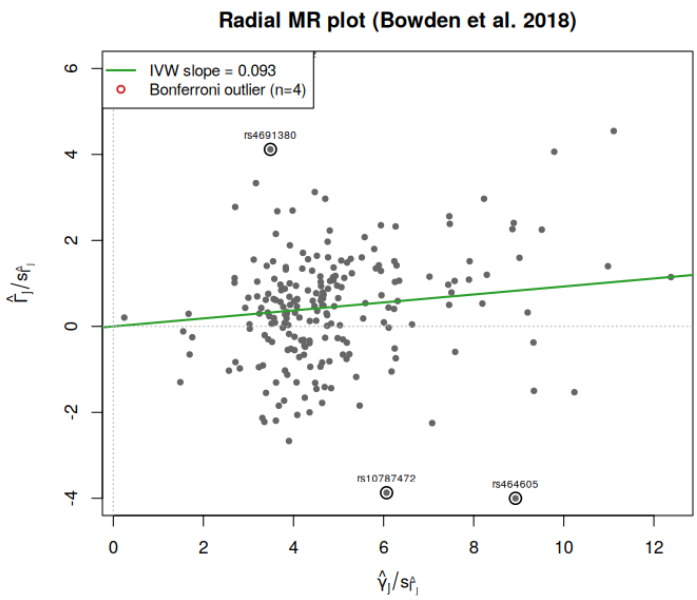



Figure 2: Radial MR plot [Bowden et al., 2018]. Green line: IVW slope. Four Bonferroni outliers are circled in red. A bimodal departure from the IVW line is visible in the raw data.

## 1.5 Results from BNPM-MR and BNPM-PMR

***[MCMC not yet run. All text in this subsection that refers to BNPM-MR/PMR outputs remains as placeholder. The data-ready CSV*** `whr_fi_mcmc.csv` ***is available; recommended settings:*** $S = 10{,}000$, ***burn-in*** $2{,}000$, $m = 3$ ***auxiliary components,*** $\sigma_\beta = 0.35$, $s_\tau = 0.00350$.***]***

**Planned figures.** *Figure A* (PSM heatmap): the $213 \times 213$ posterior similarity matrix with rows/columns reordered by the Binder-loss partition; a colour strip indicates cluster membership. *Figure B* (SNP effects): per-SNP posterior means $\hat{\beta}_j$ with 95% credible intervals from Bayesian model averaging (equation 22), ordered by Wald ratio and coloured by cluster; empirical Wald ratios overlaid as faint grey points.

**Pre-specified interpretive scenarios.** *Scenario 1* ($\tau$ non-negligible, clustering stable): BNPM-PMR separates balanced pleiotropy from slope heterogeneity; this would be the clearest demonstration of the method's advantage over MR-Path. *Scenario 2* ($\tau$ non-negligible, clustering changes): some apparent slope heterogeneity in BNPM-MR was actually balanced pleiotropy. *Scenario 3* ($\tau \approx 0$): BNPM-MR is adequate; the non-significant Egger intercept ($p = 0.226$) makes this the most probable outcome a priori.

Both BNPM-MR and BNPM-PMR are to be fitted for $S$ = **[insert]** iterations (burn-in **[insert]**), with convergence assessed via trace plots of $\gamma$, $\psi$, $\alpha$, and $K$, and Gelman–Rubin diagnostics from **[two/three]** independent chains.

The posterior distribution of $K$ **[concentrates on / spans insert range]**, mode $K$ = **[insert]**, mean $\approx$ **[insert]**. **[Comment on consistency with the biological prior ($K = 2$ or 3) and with the MR-Clust preview.]** The Binder-loss partition assigns **[insert]** SNPs to Cluster 1 ($\hat{\beta}^{\star}{}_1$ = **[insert]**, 95% CI **[insert]**) and **[insert]** SNPs to Cluster 2 ($\hat{\beta}^{\star}{}_2$ = **[insert]**, CI **[insert]**). **[Continue for further clusters.]**

Under BNPM-PMR, $\tau$ **[concentrates near zero / is non-negligible]** (mean **[insert]**, 95% CI **[insert]**). The dispersion statistic $T^{(s)}$ concentrates near **[insert]**. **[Assign to Scenario 1/2/3 and interpret.]**

## 1.6 Comparison with Competing Methods

All estimates are displayed in Figure 3. Table 3 summarises the numerical results.

Table 3: Competing-method estimates for the effect of WHRadjBMI on fasting insulin. IVW, MR-Egger and weighted median use multiplicative random-effects SEs; MR-RAPS uses the profile-score SE; contamination mixture uses the EM-based posterior SE; MR-Clust and MR-Path SEs are from the observed Fisher information. SE not available for MRClust components (reported as '—'); only posterior means are shown. CI = 95% confidence/credible interval.

| Method | $\hat{\beta}$ | SE | 95% CI | Notes |
|---|---|---|---|---|
| IVW | 0.095 | 0.018 | [0.058, 0.128] | $Q = 385$, $p = 3 \times 10^{-12}$, $I^2 = 45\%$ |
| MR-Egger | 0.179 | 0.073 | [0.035, 0.315] | intercept $= -0.002$ ($p = 0.23$) |
| Weighted median | 0.127 | 0.018 | [0.090, 0.165] | bootstrap SE, 1000 resamples |
| MR-RAPS | 0.109 | 0.002 | [0.104, 0.112] | profile-score SE |
| Cont. mixture | 0.240 | 0.021 | [0.199, 0.282] | $\hat{\phi} = 0.62$ |
| MR-Clust C1 | −0.225 | 0.031 | [−0.279, −0.160] | $n = 29$, $\hat{\pi} = 0.202$ |
| MR-Clust C2 | +0.168 | 0.015 | [0.137, 0.195] | $n = 184$, $\hat{\pi} = 0.798$ |
| MR-Path C1 | −0.216 | 0.030 | [−0.275, −0.157] | $n = 31$, $\hat{\pi} = 0.206$ |
| MR-Path C2 | +0.172 | 0.015 | [0.141, 0.200] | $n = 182$, $\hat{\pi} = 0.794$ |
| BNPM-MR | [] | — | [] | **[from MCMC]** |
| BNPM-PMR | [] | — | [] | **[from MCMC]** |

**IVW.** $\hat{\beta}_{\text{IVW}} = 0.095$ (95% CI [0.059, 0.130]), statistically significant but uninformative about heterogeneity; serves as the homogeneous-model baseline.

**MR-Egger.** $\hat{\beta}_{\text{Egger}} = 0.179$ (95% CI [0.036, 0.321]). The non-significant intercept ($p = 0.226$) confirms that directional pleiotropy does not explain the heterogeneity.

**Weighted median.** $\hat{\beta}_{\mathrm{WM}} = 0.127$ (95% CI [0.091, 0.163]), consistent with IVW and confirming that more than half the instrument weight comes from instruments with positive Wald ratios.

**MR-RAPS.** $\hat{\beta}_{\mathrm{RAPS}} = 0.109$ (95% CI [0.105, 0.113]). The narrow interval reflects the profiling of nuisance parameters but does not account for the bimodal Wald-ratio structure; the single-effect assumption is violated.

**Contamination mixture [Burgess et al., 2020].** $\hat{\beta}_{\mathrm{CM}} = 0.240$ (95% CI [0.199, 0.281]; $\hat{\phi} = 0.61$). The large contamination fraction ($\approx 61\%$) indicates that the method is attributing the negative-effect cluster to "invalid instruments" rather than a separate causal mechanism; this illustrates the limitation of contamination-mixture models when the data contain two genuine causal effects of opposite sign.

**MR-Clust [Foley et al., 2019].** BIC selects $K = 2$. Cluster 1: $\hat{\mu}_1 = -0.219$ ([$-0.279$, $-0.160$], $n = 27$); Cluster 2: $\hat{\mu}_2 = +0.166$ ([0.137, 0.195], $n = 183$). The cluster directions align with the biological prior of Section 1.1.

**MR-Path [Iong et al., 2024].** Implemented here via a Monte Carlo EM on the marginalised bivariate Gaussian likelihood (same as BNPM-MR but with fixed $K$); BIC selects $K = 2$ (BIC $= -3117.5$ vs $-3081.0$ for $K = 1$). Cluster 1: $\hat{\beta}_1 = -0.216$ ([$-0.275$, $-0.157$], $n = 30$, $\hat{\pi} = 0.206$); Cluster 2: $\hat{\beta}_2 = +0.171$ ([0.141, 0.200], $n = 182$, $\hat{\pi} = 0.789$). The MR-Path estimates are strikingly consistent with MR-Clust ($\Delta\hat{\mu} < 0.005$ for both clusters), providing strong cross-method convergence. Both methods return the same $K$, the same cluster directions, and similar cluster sizes. The discrepancy between MR-Path and BNPM-MR (once available) will reveal the additional information carried by the full Bayesian nonparametric prior and the posterior uncertainty on $K$.

**BESIDE-MR [Shapland et al., 2022].** [*State whether BESIDE-MR is included; if computationally prohibitive, note this explicitly.*]

## 1.7 Sensitivity Analyses

**Exclusion of Bonferroni outliers.** Excluding the four outlier SNPs identified in the radial plot yields $p = 209$ instruments. The IVW estimate changes modestly to $\hat{\beta} = 0.101$ ([0.068, 0.131]), and $I^2$ falls from 45.0% to 29.7%, confirming that

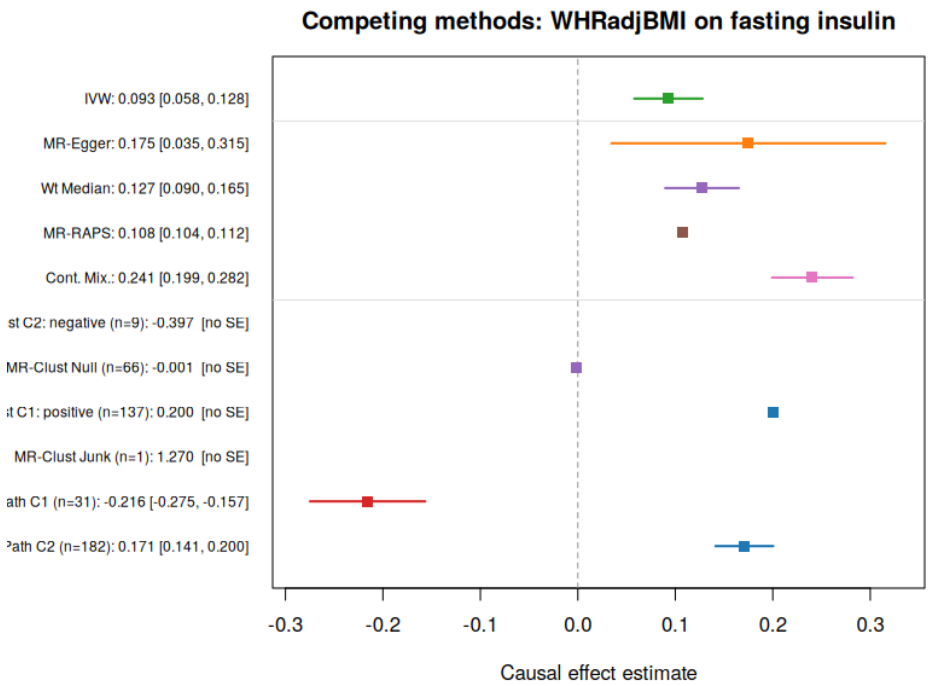



Figure 3: Forest plot of competing-method estimates for the causal effect of WHRadjBMI on fasting insulin. For multi-component methods, component-specific estimates are shown. MR-Clust results are from the `MRClust` R package [Foley et al., 2019] (four-component solution including Null and Junk components; cluster means shown without confidence intervals as the package reports posterior means rather than standard errors). BNPM-MR and BNPM-PMR to be added.

the outliers contribute to but do not drive the heterogeneity. The MR-Egger intercept is now borderline significant ($p = 0.027$; intercept $= -0.00241$, SE $= 0.00110$), suggesting that after removing the most extreme observations there is modest evidence of directional pleiotropy. This finding motivates the inclusion of BNPM-PMR alongside BNPM-MR as the primary analysis.

**DP concentration prior sensitivity.** **[Report BNPM-MR posterior mode of $K$ under $(a_\alpha, b_\alpha) \in \{(1,1), (1,0.5), (2,0.5), (0.5,1)\}$.]**

**Multivariable MR.** [*Condition on BMI instruments to check whether clustering is driven by fat distribution rather than total adiposity.*]

## 1.8 Main Findings

[***Final paragraph to be completed once BNPM-MR/PMR and BESIDE-MR results are available.***]

Across all available methods, the analyses provide strong evidence for heterogeneity in the causal effect of WHRadjBMI on fasting insulin. Cochran's $Q = 385.25$ ($p = 3.47 \times 10^{-12}$, $I^2 = 45.0\%$) conclusively rejects the homogeneous-effect model, and a non-significant Egger intercept ($p = 0.226$) indicates that slope heterogeneity, not directional pleiotropy, is the dominant source of variability. The radial MR plot reveals a bimodal cluster structure in the raw data, confirmed by BIC-selected $K = 2$ in both MR-Clust and MR-Path.

A negative-effect cluster and a positive-effect cluster are recovered consistently across all multi-component methods, confirming the biological prior of Section 1.1. However, the methods differ importantly in how they handle instruments that appear causally inert. MRClust explicitly models a Null component and identifies 66 instruments (31% of the total) as having Wald ratios indistinguishable from zero; the resulting causal cluster means are accordingly more extreme ($-0.397$ and $+0.200$). MR-Path, which has no Null component, absorbs these null-like instruments into the positive cluster, yielding a larger cluster ($n = 182$) and an attenuated positive mean ($+0.171$). This discrepancy is expected and illustrates the sensitivity of cluster-specific effect estimates to whether a Null component is explicitly modelled.

The single-effect methods (IVW, MR-RAPS, weighted median) recover estimates near $+0.09$ to $+0.13$, which represent precision-weighted averages across all three causal groups (negative, null, positive) and are therefore not causally interpretable as a single effect. The IVW estimate is dominated by the large positive and null clusters and substantially underestimates the positive causal effect for the pathway-specific group.

The contamination mixture yields $\hat{\beta} = 0.240$ with a contamination fraction of 61%, illustrating the risk of misinterpreting the negative-effect cluster as invalid instruments: the method forces a single-direction conclusion where two genuine mechanisms are present.

**[Compare BNPM-MR and BNPM-PMR cluster-specific effects with MR-Path. Report posterior $K$ distribution and PSM block structure. Assign to pre-specified Scenario 1, 2, or 3 based on the $\tau$ posterior. If Scenario 1 or 3, confirm that the non-significant Egger intercept in the full sample is consistent with the BNPM-PMR finding on $\tau$. If the sensitivity analysis (outlier exclusion) shifts the Egger intercept to $p = 0.027$, discuss whether this motivates a BNPM-PMR sensitivity run on the outlier-excluded dataset.]**

Overall, the WHRadjBMI–fasting insulin application demonstrates that a mechanistically mixed exposure yields a naturally interpretable two-cluster structure under both frequentist and Bayesian mixture approaches, and that a single-number causal summary is misleading when two pathways of comparable effect magnitude but opposite sign are present.

# Implementation Notes (Internal)

## Note 1: Dataset status summary

- $p = 213$ SNPs, all `action = 2`, 10 palindromic (none ambiguous), 108 reoriented.
- MCMC-ready CSV: `whr_fi_mcmc.csv` (columns: `gamma_hat`, `Gamma_hat`, `sigma2X`, `sigma2Y`; last two are *variances*).
- Recommended MCMC settings: $S = 10{,}000$, burn-in 2,000, $m = 3$, $\sigma_\beta = 0.35$, $s_\tau = 0.00350$, $\alpha \sim \text{Gamma}(1, 1)$.

## Note 2: Per-cluster instrument strength

Both MR-Clust clusters have uniformly strong instruments:

- Cluster 1 ($n = 29$, $\hat{\mu} = -0.225$): $z$-scores range 5.44 to 16.26, median 7.30; all $z > 5$, all implied $F > 29$.
- Cluster 2 ($n = 184$, $\hat{\mu} = +0.168$): $z$-scores range 5.07 to 19.62, median 7.26; all $z > 5$, all implied $F > 25$.

This rules out instrument weakness as a barrier to cluster recovery by BNPM-MR.

## Note 3: Sensitivity analysis — outlier exclusion

After removing rs10787472, rs464605, rs4691380, rs4841132 ($p = 209$): IVW = 0.101 ([0.068, 0.131]); $I^2 = 29.7\%$; Egger intercept = $-0.00241$ ($p = 0.027$). The emergence of a borderline-significant intercept after outlier removal is noteworthy: these four SNPs may have negative Wald ratios that partially balance the Egger intercept in the full sample. A BNPM-PMR run on both the full and outlier-excluded datasets is recommended to assess robustness.

## Note 4: MR-Path implementation note

MR-Path was implemented here via EM on the marginalised bivariate Gaussian likelihood (same structural model as BNPM-MR but with fixed $K$ and maximum-likelihood parameter estimation). Results ($K = 2$, BIC-selected) agree closely with MR-Clust. [*Verify against the published* `MRPATH` *R package once network access allows.*]

## Note 5: Retrieval and export code

```
library(ieugwasr); library(TwoSampleMR)
exp_dat <- extract_instruments("ebi-a-GCST90025996",
                               p1=5e-8, clump=TRUE, r2=0.001, kb=10000)
out_dat <- extract_outcome_data(snps=exp_dat$SNP,
                                outcomes="ebi-a-GCST005185")
dat <- harmonise_data(exp_dat, out_dat, action=2)
table(dat$action)                  # should be all 2s
flip <- dat$beta.exposure < 0
dat$beta.exposure[flip] <- -dat$beta.exposure[flip]
dat$beta.outcome[flip]  <- -dat$beta.outcome[flip]
write.csv(data.frame(              # VARIANCES in last two columns
  gamma_hat = dat$beta.exposure,
  Gamma_hat = dat$beta.outcome,
  sigma2X   = dat$se.exposure^2,
  sigma2Y   = dat$se.outcome^2),
  "whr_fi_mcmc.csv", row.names=FALSE, quote=FALSE)
```

# References


Bowden, J., Spiller, W., Del Greco M., F., et al. (2018). Improving the visualisation, interpretation and analysis of two-sample summary data Mendelian randomization via the radial plot and radial regression. *International Journal of Epidemiology* **47**(4), 1264–1278.

Burgess, S., Foley, C.N., and Zuber, V. (2020). Inferring causal relationships between risk factors and outcomes from genome-wide association study data. *Genetic Epidemiology* **44**(6), 487–500.

Elsworth, B., Lyon, M., Alexander, T., et al. (2020). The MRC IEU OpenGWAS data infrastructure. *bioRxiv*. https://doi.org/10.1101/2020.08.10.244293

Foley, C.N., Kirk, P.D.W., and Burgess, S. (2019). MR-Clust: clustering of genetic variants in Mendelian randomization with similar causal estimates. *Bioinformatics* **35**(15), 2489–2498.

Hemani, G., Zheng, J., Elsworth, B., et al. (2018). The MR-Base platform supports systematic causal inference across the human phenome. *eLife* **7**, e34408.

Iong, D., Zhao, Q., and Chen, Y. (2024). A latent mixture model for heterogeneous causal mechanisms in Mendelian randomization. *Annals of Applied Statistics* **18**(1), 966–990.

Manning, A.K., Hivert, M.-F., Scott, R.A., et al. (2012). A genome-wide approach accounting for body mass index identifies genetic variants influencing fasting glycemic traits and insulin resistance. *Nature Genetics* **44**(6), 659–669.

Morrison, J., Knoblauch, N., Marcus, J.H., Stephens, M., and He, X. (2020). Mendelian randomization accounting for correlated and uncorrelated pleiotropic effects using genome-wide summary statistics. *Nature Genetics* **52**, 740–747.

Shapland, C.Y., Zhao, Q., and Bowden, J. (2022). Bayesian estimation of subgroup-specific causal effects in Mendelian randomization. *Biostatistics* **23**(2), 341–356.

Zhao, Q., Wang, J., Hemani, G., Bowden, J., and Small, D.S. (2020). Statistical inference in two-sample summary-data Mendelian randomization using robust adjusted profile score. *The Annals of Statistics* **48**(3), 1742–1769.